

Projet minilab

configuration réseau sur le serveur principal

```
GNU nano 8.4 /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto ens33
iface ens33 inet dhcp

auto ens34
iface ens34 inet static
    address 192.168.15.254/24
```

Préparation du RAID 5

installation de mdadm

```
user@minilab:~$ sudo apt install mdadm -y
Installation de :
  mdadm

Paquets suggérés :
  default-mta | mail-transport-agent

Sommaire :
  Mise à niveau de : 0. Installation de : 1 Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 444 kB
Espace nécessaire : 1 310 kB / 2 561 MB disponible

Réception de : 1 http://deb.debian.org/debian trixie/main amd64 mdadm amd64 4.4-11 [444 kB]
444 ko réceptionnés en 0s (6 194 ko/s)
Préconfiguration des paquets...
Sélection du paquet mdadm précédemment désélectionné.
(Lecture de la base de données... 168159 fichiers et répertoires déjà installés.
)
Préparation du dépaquetage de .../mdadm_4.4-11_amd64.deb ...
Dépaquetage de mdadm (4.4-11) ...
Paramétrage de mdadm (4.4-11) ...
Generating mdadm.conf... done.
update-initramfs: deferring update (trigger activated)
Generating grub configuration file ...
Found background image: /usr/share/images/desktop-base/desktop-grub.png
Found linux image: /boot/vmlinuz-6.12.90+deb13.1-amd64
Found initrd image: /boot/initrd.img-6.12.90+deb13.1-amd64
Found linux image: /boot/vmlinuz-6.12.57+deb13-amd64
Found initrd image: /boot/initrd.img-6.12.57+deb13-amd64
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
```

vérification des disque disponible et création du RAID

```
user@minilab:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0    8G  0 disk
├─sda1      8:1    0   7,5G  0 part /
├─sda2      8:2    0    1K  0 part
└─sda5      8:5    0   480M  0 part [SWAP]
sdb         8:16   0    8G  0 disk
sdc         8:32   0    8G  0 disk
sdd         8:48   0    8G  0 disk
sde         8:64   0    8G  0 disk
sr0         11:0    1 1024M  0 rom
user@minilab:~$ mdadm --create --verbose /dev/md0 --level=5 --raid-devices=4 /dev/sdb /dev/sdc /dev/sdd /dev/sde
bash: mdadm : commande introuvable
user@minilab:~$ sudo mdadm --create --verbose /dev/md0 --level=5 --raid-devices=4 /dev/sdb /dev/sdc /dev/sdd /dev/sde
To optimize recovery speed, it is recommended to enable write-indent bitmap, do you want to enable it now? [y/N]? y
```

formatage des disque du RAID en "ext4"

```
user@minilab:~$ sudo mkfs.ext4 /dev/md0
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 6284544 4k blocks and 1572864 inodes
Filesystem UUID: 0dfd01e7-27fa-4852-9e19-4aca20ed0b81
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

user@minilab:~$
```

modification du fichier :etc/fstab pour pouvoir démarrer sur le raid au démarrage

```
GNU nano 8.4 /etc/fstab *
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda1 during installation
UUID=5ef18d0d-5e10-4247-9f66-9c87297761f5 / ext4 errors=remount-ro 0 1
# swap was on /dev/sda5 during installation
UUID=9b2acf33-8153-4a04-8b5f-7c7021588fa6 none swap sw 0 0
/dev/sr0 /media/cdrom0 udf,iso9660 user,noauto 0 0
/dev/md0 /home ext4 default 0 2
```

(note: debian a renommer le raid donc j'ai remplacer "/dev/md0" par son UUID pour etre sur qu'il ne sera pas renommer)

```
raid@serveurRAID:~$ sudo mdadm --detail --scan | sudo tee -a /etc/mdadm/mdadm.conf
[sudo] Mot de passe de raid :
ARRAY /dev/md/serveurRAID:0 metadata=1.2 UUID=73aacd5b:795669a1:a88de160:928fdda4
raid@serveurRAID:~$ sudo update-initramfs -u
update-initramfs: Generating /boot/initrd.img-6.12.90+deb13.1-amd64
raid@serveurRAID:~$ sudo blkid | grep md
/dev/md127: UUID="1f35da8a-862e-4747-887a-a424f3676c19" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sr0: BLOCK_SIZE="2048" UUID="2025-11-15-11-09-09-00" LABEL="Debian 13.2.0 a md64 n" TYPE="iso9660" PTUUID="4a641575" PTTYPE="dos"
raid@serveurRAID:~$ sudo nano /etc/fstab
```

```
GNU nano 8.4 /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda1 during installation
UUID=a3025a1e-8f69-46c2-81a0-8527dd7afa52 / ext4 errors=remount>
# swap was on /dev/sda5 during installation
UUID=ccd6df3d-fcca-442f-b64c-b5707402bd5f none swap sw
/dev/sr0 /media/cdrom0 udf,iso9660 user,noauto 0 0
UUID=1f35da8a-862e-4747-887a-a424f3676c19 /home ext4 defaults
```

installation du serveur NFS

```
user@minilab:~$ sudo apt install nfs-kernel-server
Installation de :
  nfs-kernel-server

Installation de dépendances :
  keyutils libevent-core-2.1-7t64 libnfsidmap1 nfs-common python3-yaml rpcbind

Paquets suggérés :
  open-iscsi watchdog

Sommaire :
  Mise à niveau de : 0. Installation de : 7Supprimé : 0. Non mis à jour : 0
  Taille du téléchargement : 890 kB
  Espace nécessaire : 3 480 kB / 2 558 MB disponible

Continuer ? [O/n]
```

déclaration du partage dans le fichier "/etc/exports" (L'option `no_root_squash` est très utile ici pour la gestion des profils itinérants, car elle permet au serveur de conserver les pleins droits lors de la création automatique des dossiers personnels des nouveaux utilisateurs)

```

GNU nano 8.4 /etc/exports
# /etc/exports: the access control list for filesystems which may be exported
# to NFS clients. See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_sub>
#
# Example for NFSv4:
# /srv/nfs4 gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes gss/krb5i(rw,sync,no_subtree_check)
#
/home 192.168.15.0/24(rw,sync,no_subtree_check,no_root_squash)

```

redémarrage du service et vérification

```

raid@serveurRAID:~$ sudo exportfs -arv
exporting 192.168.15.0/24:/home
raid@serveurRAID:~$ sudo systemctl restart nfs-kernel-server
raid@serveurRAID:~$

```

```

raid@serveurRAID:~$ sudo showmount -e 127.0.0.1
Export list for 127.0.0.1:
/home 192.168.15.0/24
raid@serveurRAID:~$

```

VM master/slave

configuration des adresse ip

dhcp 1(master): 192.168.15.253

dhcp 2(slave): 192.168.15.252

```

GNU nano 8.4 /etc/network/interfaces *
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto ens33
iface ens33 inet static
    address 192.168.15.253/24

```

installation de keepalived sur les 2VM pour mettre en place le service "failover"

```

master@ServerMaster:~$ sudo apt install keepalived
[sudo] Mot de passe de master :
Installation de :
  keepalived

Installation de dépendances :
  ipvsadm

Paquets suggérés :
  heartbeat ldirectord

Sommaire :
  Mise à niveau de : 0. Installation de : 2Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 632 kB
Espace nécessaire : 1 576 kB / 2 526 MB disponible

Continuer ? [O/n]

```

configuration du failover sur le master (remplacer state MASTER par state BACKUP sur le slave)

```

GNU nano 8.4 /etc/keepalived/keepalived.conf *
vrrp_instance VI_1 {
    state MASTER
    interface ens33
    virtual_router_id 51
    priority 100
    advert_int 1
    authentication {
        auth_type PASS
        auth_pass laplateforme
    }
    virtual_ipaddress {
        192.168.15.250/24
    }
}

```

résultat sur la vm Master

```

2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:e5:f0:1f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29e5f01f
    inet 192.168.15.253/24 brd 192.168.15.255 scope global ens33
        valid_lft forever preferred_lft forever
    inet 192.168.15.250/24 scope global secondary proto 0x12 ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fee5:f01f/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
master@ServerMaster:~$

```

installation du serveur dhcp sur les 2 VM

```

master@ServerMaster:~$ sudo apt install isc-dhcp-server
[sudo] Mot de passe de master :
Installation de :
  isc-dhcp-server

Installation de dépendances :
  isc-dhcp-common policycoreutils selinux-utils

Paquets suggérés :
  isc-dhcp-server-ldap ieee-data

Sommaire :
  Mise à niveau de : 0. Installation de : 4 Supprimé : 0. Non mis à jour :
  0
Taille du téléchargement : 1 832 kB
Espace nécessaire : 8 002 kB / 2 513 MB disponible

Continuer ? [0/n]

```

ajout de l'interface réseau

```

GNU nano 8.4 /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID ins
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens33"
INTERFACESv6=""

```

configuration du dhcp sur le master

```

# Déclaration du Failover (Mode Primaire)
failover peer "dhcp-failover" {
    primary;
    address 192.168.15.253;
    port 519;
    peer address 192.168.15.252;
    peer port 520;
    max-response-delay 60;
    max-unacked-updates 10;
    mclt 3600;
    split 128;
    load balance max seconds 3;
}

# Configuration du réseau et des options
subnet 192.168.15.0 netmask 255.255.255.0 {
    option routers 192.168.15.254;
    option domain-name "linuxisgood.local";
    option domain-name-servers 192.168.15.250;

    # Pool d'adresses synchronisé
    pool {
        failover peer "dhcp-failover";
        range 192.168.15.100 192.168.15.150;
    }
}

```

sur le slave

```
GNU nano 8.4 /etc/dhcp/dhcpd.conf *
# Déclaration du Failover (Mode Secondaire)
failover peer "dhcp-failover" {
    secondary;
    address 192.168.15.252;
    port 520;
    peer address 192.168.15.253;
    peer port 519;
    max-response-delay 60;
    max-unacked-updates 10;
    load balance max seconds 3;
}

# Configuration du réseau et des options
subnet 192.168.15.0 netmask 255.255.255.0 {
    option routers 192.168.15.254;
    option domain-name "linuxisgood.local";
    option domain-name-servers 192.168.15.250;

    pool {
        failover peer "dhcp-failover";
        range 192.168.15.100 192.168.15.150;
    }
}
```

test en regardant les log

```
master@ServerMaster:~$ sudo journalctl -u isc-dhcp-server -f
juin 17 09:26:55 ServerMaster dhcpd[2282]: failover peer dhcp-failover: I move from recover to recover-done
juin 17 09:26:55 ServerMaster dhcpd[2282]: Sent update done message to dhcp-failover
juin 17 09:26:55 ServerMaster dhcpd[2282]: failover peer dhcp-failover: peer moves from recover to recover-done
juin 17 09:26:55 ServerMaster dhcpd[2282]: Both servers have entered recover-done!
juin 17 09:26:55 ServerMaster dhcpd[2282]: failover peer dhcp-failover: I move from recover-done to normal
juin 17 09:26:55 ServerMaster dhcpd[2282]: balancing pool 560dab159c50 192.168.15.0/24 total 51 free 51 backup 0 lts 25 max-
juin 17 09:26:55 ServerMaster dhcpd[2282]: balanced pool 560dab159c50 192.168.15.0/24 total 51 free 26 backup 25 lts 0 max-m
juin 17 09:26:55 ServerMaster dhcpd[2282]: Sending updates to dhcp-failover.
juin 17 09:26:55 ServerMaster dhcpd[2282]: failover peer dhcp-failover: peer moves from recover-done to normal
juin 17 09:26:55 ServerMaster dhcpd[2282]: failover peer dhcp-failover: Both servers normal
```

installation du service DNS

```
slave@ServerSlave:~$ sudo apt install bind9 bind9utils bind9-doc dnsutils
[sudo] Mot de passe de slave :
Note : sélection de « bind9-utils » au lieu de « bind9utils »
Note : sélection de « bind9-dnsutils » au lieu de « dnsutils »
bind9-dnsutils est déjà la version la plus récente (1:9.20.23-1~deb13u1).
Installation de :
  bind9 bind9-doc bind9-utils

Paquets suggérés :
  resolvconf ufw

Sommaire :
  Mise à niveau de : 0. Installation de : 3Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 4 053 kB
  Espace nécessaire : 9 506 kB / 2 549 MB disponible

Continuer ? [O/n]
```


configuration du DNS sur le master

```
GNU nano 8.4 /etc/bind/named.conf.local
//
// Do any local configuration here
//
zone "linuxisgood.local" {
    type master;
    file "/etc/bind/db.linuxisgood.local";
    allow-transfer { 192.168.15.252; }; # Autorise le Slave à copier la zone
    also-notify { 192.168.15.252; };    # Préviens le Slave en cas de changement
};
```

```
GNU nano 8.4 /etc/bind/db.linuxisgood.local
$TTL      604800
@         IN      SOA      ns1.linuxisgood.local. admin.linuxisgood.local. (
                                2           ; Serial
                                604800       ; Refresh
                                86400        ; Retry
                                2419200     ; Expire
                                604800 )    ; Negative Cache TTL
;
@         IN      NS       ns1.linuxisgood.local.
@         IN      NS       ns2.linuxisgood.local.

ns1       IN      A        192.168.15.253
ns2       IN      A        192.168.15.252
router    IN      A        192.168.15.254
```

sur le slave

```
GNU nano 8.4 /etc/bind/named.conf
//
// Do any local configuration here
//
zone "linuxisgood.local" {
    type slave;
    file "/var/cache/bind/db.linuxisgood.local"; # Dossier où le Slave a le droit d'écrire
    masters { 192.168.15.253; };
};
```

installation de ldap

```
master@ServerMaster:~$ sudo apt install slapd ldap-utils
Installation de :
  ldap-utils  slapd

Installation de dépendances :
  libodbc2  libodbcrc2

Paquets suggérés :
  libsassl2-modules-gssapi-mit | libsassl2-modules-gssapi-heimdal  odbc-postgresql  tdsodbc

Sommaire :
  Mise à niveau de : 0. Installation de : 4Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 1 795 kB
Espace nécessaire : 6 207 kB / 2 495 MB disponible

Continuer ? [O/n]
```


configuration du provider pour la réplication (/etc/provider.ldif)

```
GNU nano 8.4 provider.ldif
dn: cn=module{0},cn=config
changetype: modify
add: olcModuleLoad
olcModuleLoad: syncprov

dn: olcOverlay=syncprov,olcDatabase={1}mdb,cn=config
changetype: add
objectClass: olcOverlayConfig
objectClass: olcSyncProvConfig
olcOverlay: syncprov
olcSpCheckpoint: 100 10
olcSpSessionlog: 100
```

configuration du consumer

```
GNU nano 8.4 consumer.ldif *
dn: olcDatabase={1}mdb,cn=config
changetype: modify
add: olcSyncRepl
olcSyncRepl: rid=001 provider=ldap://192.168.15.253 bindmethod=simple binddn="cn=admin,dc=linuxisgood,dc=local" credentials=root searchbase="dc=linuxisgood,dc=local" ty
```

envoi des config au serveur ldap

```
slave@ServerSlave:~$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f consumer.ldif
modifying entry "olcDatabase={1}mdb,cn=config"

slave@ServerSlave:~$
```

installation de ldapadmin

```
master@ServerMaster:~$ sudo apt install phpldapadmin
Installation de :
  phpldapadmin

Installation de dépendances :
  libapache2-mod-php8.4 php php-common php-ldap php-xml php8.4 php8.4-cli php8.4-common php8.4-ldap php8.4-opcache php8.4-readline php8.4-xml

Paquets suggérés :
  php-pear

Sommaire :
  Mise à niveau de : 0. Installation de : 13Supprimé : 0. Non mis à jour : 0
  Taille du téléchargement : 6 355 kB
  Espace nécessaire : 29,8 MB / 2 488 MB disponible

Continuer ? [O/n]
```

configuration

```
/* Array of base DN's of your LDAP server. Leave this blank to have phpldapadmin
   auto-detect it for you. */
$servers->setValue('server','base',array('dc=linuxisgood,dc=local'));

$servers->setValue('login','bind_id','cn=admin,dc=linuxisgood,dc=local');
# $servers->setValue('login','bind_id','cn=Manager,dc=linuxisgood,dc=local');
```

création des groupe + utilisateur depuis le GUI

The screenshot shows a user creation interface with the following fields and values:

- Username:** client1
- Nom Commun:** client1
- ID utilisateur:** client1
- Mot de passe:** md5 (with a confirmation field)
- UID:** 1001
- GID:** groupe
- Répertoire personnel:** /home/client1
- Shell de connexion:** Bash

depuis le serveur nfs, configuration du routage pour que les vm host-only puissent accéder à internet

```
GNU nano 8.4 /etc/sysctl.conf
net.ipv4.ip_forward=1
```

application à l'interface réseau nat

```
root@serveurRAID:~# iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE
root@serveurRAID:~#
```

```
root@client:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=127 time=6.33 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=127 time=14.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=127 time=42.9 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 6.329/21.242/42.878/15.659 ms
root@client:~#
```

installation du client ldap

```

root@client:~# apt install libnss-ldapd libpam-ldapd nslcd
Installation de :
  libnss-ldapd libpam-ldapd nslcd

Installation de dépendances :
  ldap-utils nscd nslcd-utils

Paquets suggérés :
  libsasl2-modules-gssapi-mit | libsasl2-modules-gssapi-heimdal kstart

Sommaire :
  Mise à niveau de : 0. Installation de : 6Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 555 kB
Espace nécessaire : 1 817 kB / 10,2 GB disponible

Continuer ? [0/n] █

```

configuration

```

GNU nano 8.4 /etc/nsswitch.conf
# /etc/nsswitch.conf
#
# Example configuration of GNU Name Service Switch f
# If you have the `glibc-doc-reference' and `info' p
# `info libc "Name Service Switch"' for information

passwd:          files systemd ldap
group:           files systemd ldap
shadow:          files ldap
gshadow:         files systemd

```

test vérification

```

root@serveurRAID:~# getent passwd client1
client1:*:1001:500: client1:/home/client1:/bin/bash
root@serveurRAID:~#

```

test connection

```

user@client:~$ su client1
Mot de passe :
client1@client:/home/user$ █

```

```

GNU nano 8.4 /etc/exports
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes      hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4       gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes gss/krb5i(rw,sync,no_subtree_check)
#
/home 192.168.15.0/24(rw,sync,no_subtree_check,no_root_squash)
/var/lib/google-authenticator 192.168.15.0/24(rw,sync,no_root_squash,fsid=0)

```

installation du vpn + configuration et création du fichier ovpn

```

root@serveurRAID:~# chmod +x openvpn-install.sh
root@serveurRAID:~# ./openvpn-install.sh
OpenVPN installer and manager

Usage: openvpn-install <command> [options]

Commands:
install      Install and configure OpenVPN server
uninstall    Remove OpenVPN server
client       Manage client certificates
server       Server management
interactive   Launch interactive menu

Global Options:
--verbose    Show detailed output
--log <path> Log file path (default: openvpn-install.log)
--no-log     Disable file logging
--no-color   Disable colored output
-h, --help   Show help

Run 'openvpn-install <command> --help' for command-specific help.
root@serveurRAID:~# interactive

```

installation du pare-feu iptable

```

root@serveurRAID:~# apt install iptables-persistent
Installation de :
  iptables-persistent

Installation de dépendances :
  netfilter-persistent

Sommaire :
  Mise à niveau de : 0. Installation de : 2Supprimé : 0. Non mis à jour : 6
Taille du téléchargement : 18,5 kB
Espace nécessaire : 96,3 kB / 2 535 MB disponible

Continuer ? [0/n] █

```

configuration en bloquant toute connection extérieur sauf vpn

```
GNU nano 8.4 /root/firewall.sh
#!/bin/bash

WAN="ens33"
LAN="ens34"
sysctl -w net.ipv4.ip_forward=1

# =====
# 1. NETTOYAGE
# =====
iptables -F
iptables -X
iptables -t nat -F
iptables -t nat -X

# =====
# 2. POLITIQUES PAR DÉFAUT (Sécurité stricte)
# =====
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT

# =====
# 3. RÈGLES INPUT (Trafic adressé à la passerelle)
# =====
# Autoriser le trafic local interne
iptables -A INPUT -i lo -j ACCEPT
# Autoriser le retour des connexions établies (Ping, Maj...)
iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
# Autoriser le réseau local (LAN)
iptables -A INPUT -i $LAN -s 192.168.15.0/24 -j ACCEPT
```

```

iptables -A INPUT -i tun0 -j ACCEPT
# Autoriser la connexion OpenVPN depuis Internet
iptables -A INPUT -i $WAN -p udp --dport 1194 -j ACCEPT

# connection ssh
#iptables -A INPUT -i $WAN -p tcp --dport 22 -j ACCEPT

# =====
# 4. RÈGLES FORWARD (Trafic traversant la passerelle)
# =====
# Autoriser le retour des connexions établies (DNS, Web...)
iptables -A FORWARD -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
# Autoriser le transit des requêtes DNS
iptables -A FORWARD -p udp --dport 53 -j ACCEPT
iptables -A FORWARD -p tcp --dport 53 -j ACCEPT
# Autoriser le LAN à sortir sur Internet
iptables -A FORWARD -i $LAN -o $WAN -j ACCEPT
# Autoriser le VPN à communiquer avec le LAN
iptables -A FORWARD -i tun0 -o $LAN -j ACCEPT
iptables -A FORWARD -i $LAN -o tun0 -j ACCEPT

# =====
# 5. RÈGLES NAT (Traduction d'adresses)
# =====
# Donner Internet au LAN
iptables -t nat -A POSTROUTING -o $WAN -j MASQUERADE
# Donner accès au LAN depuis le VPN
iptables -t nat -A POSTROUTING -s 10.8.0.0/24 -o $LAN -j MASQUERADE

# =====
# 6. SAUVEGARDE
# =====
# On sauvegarde tout à la toute fin pour être sûr de rien oublier
netfilter-persistent save

```

donner les droit d'exécution

```

root@serveurRAID:~# chmod +x /root/firewall.sh
root@serveurRAID:~# /root/firewall.sh
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
root@serveurRAID:~# █

```

Passage de ldap en ldaps

```

GNU nano 8.4 /root/certinfo.ldif *
dn: cn=config
changetype: modify
add: olcTLSCACertificateFile
olcTLSCACertificateFile: /etc/ssl/certs/ssl-cert-snakeoil.pem
-
add: olcTLSCertificateFile
olcTLSCertificateFile: /etc/ssl/certs/ssl-cert-snakeoil.pem
-
add: olcTLSCertificateKeyFile
olcTLSCertificateKeyFile: /etc/ssl/private/ssl-cert-snakeoil.key

```

envoi au serveur

```

master@ServerMaster:~$ sudo ldapmodify -Y EXTERNAL -H ldapi:/// -f /root/
certinfo.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
modifying entry "cn=config"
ldap_modify: Other (e.g., implementation specific) error (80)

master@ServerMaster:~$

```

```

# SLAPD_SERVICES="ldap://127.0.0.1:389/ ldaps://127.0.0.1:636/
SLAPD_SERVICES="ldap:/// ldapi:/// ldaps:///"

```

installation de webmin (sur nfs pour voir l'état du raid + nsf et master pour dhcp et dns)

```

master@ServerMaster:~$ curl -o setup-repos.sh https://raw.githubusercontent.com/
webmin/webmin/master/setup-repos.sh
sh setup-repos.sh

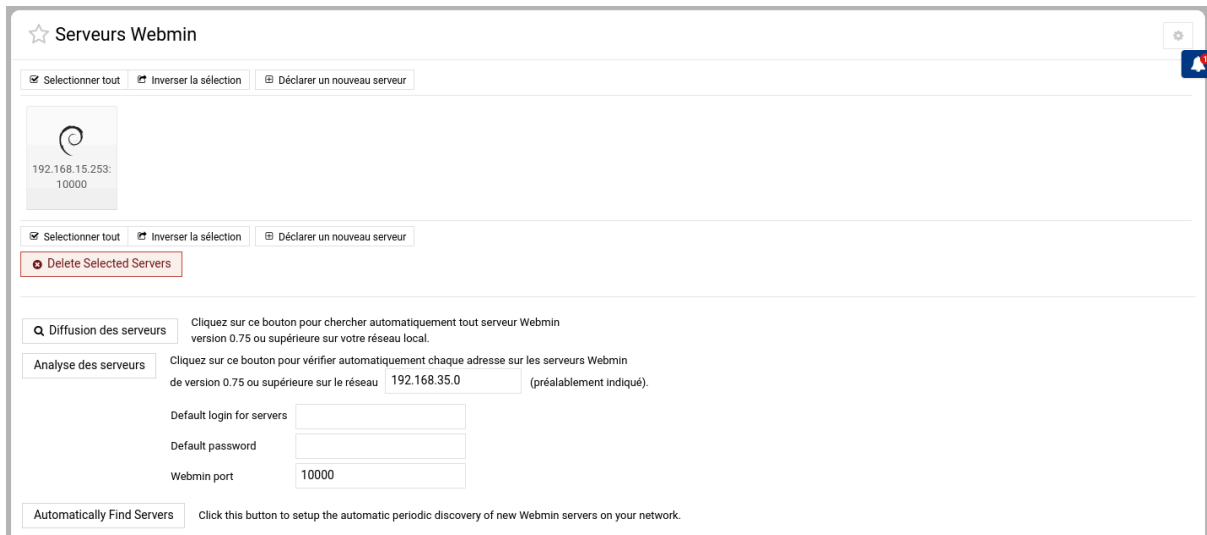
```

liaison des deux webmin en 1 seul interface

☆ Déclaration d'un serveur

Détails du serveur	
Nom d'hôte 192.168.15.253	Description ☒ Depuis le nom d'hôte et le port
Port 10000	Membre des groupes de serveurs Nouveau groupe:
Type de serveur Debian Linux	Type de lien ☒ Lien normal vers le serveur ☐ Login via Webmin with nom d'utilisat and mot de pass ☐ Se connecter quand on clique sur l'icône
Serveur SSL ☐ Oui ☒ Non ☐ Check remote SSL certificate?	
Sauvegarder	

[← Retourner à la liste des serveurs](#)



passage en https de phpldapadmin

activation du module ssl d'apache2

```

root@ServerMaster:~# a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
    systemctl restart apache2
root@ServerMaster:~# a2ensite default-ssl
Enabling site default-ssl.
To activate the new configuration, you need to run:
    systemctl reload apache2
root@ServerMaster:~# systemctl restart apache2
root@ServerMaster:~#

```

redirection automatique en https

```

GNU nano 8.4 /etc/apache2/sites-available/000-default.conf *
<VirtualHost *:80>
    Redirect permanent / https://192.168.15.253/
    # The ServerName directive sets the request scheme, hostname and port t
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

```

préparation pour le script de mise à jour

génération d'une clé ssh

```

root@serveurRAID:~# ssh-keygen -t rsa -b 4096 -N "" -f ~/.ssh/id_rsa
Generating public/private rsa key pair.
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:Bi3zki76QaW+itvq6dHx+zvG6yXDKfv7d3EZIkJp5+A root@serveurRAID
The key's randomart image is:
+---[RSA 4096]-----+
|
|      .
|     . = .
|    = = +
|   o * E o . .
|  .o o S . . . o
| .oo..o. . o
| . .+oo= . o
| oo. ++++ . .
| **+ooo=B*.. .
+----[SHA256]-----+
root@serveurRAID:~# ssh-copy-id root@192.168.15.253
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '192.168.15.253 (192.168.15.253)' can't be established.
ED25519 key fingerprint is SHA256:OpdP13IxGhGNhPNRz9LWCFdQVXYfZxnZAAuhZf

```

copie de la clef ssh a toute les vm

```

root@serveurRAID:~# ssh-copy-id root@192.168.15.100
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '192.168.15.100 (192.168.15.100)' can't be established.
ED25519 key fingerprint is SHA256:YSozRJZyD4WtWg4m2msYLv2UgC1QZT00Z0ni40wNo6Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@192.168.15.100's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@192.168.15.100'"
and check to make sure that only the key(s) you wanted were added.

root@serveurRAID:~# █

```

script de mise à jour sur toute les VM

```
#!/bin/bash

# Liste des machines à mettre à jour
SERVEURS=("192.168.15.254" "192.168.15.253" "192.168.15.252" "192.168.15.101")

echo "====="
echo "🚀 DÉBUT DE LA MISE À JOUR GLOBALE 🚀"
echo "Date : $(date)"
echo "====="

for ip in "${SERVEURS[@]}; do
    echo ""
    echo "📦 -> Connexion à la machine : $ip"

    # Si c'est la machine locale (la passerelle), on ne fait pas de SSH
    if [ "$ip" == "192.168.15.254" ] || [ "$ip" == "127.0.0.1" ]; then
        apt update && apt upgrade -y
    else
        # Exécution de la commande de mise à jour à distance via SSH
        ssh -n -o BatchMode=yes root@$ip "apt update && apt upgrade -y"
    fi

    if [ $? -eq 0 ]; then
        echo "✓ -> Mise à jour réussie pour $ip"
    else
        echo "✗ -> Erreur de mise à jour pour $ip"
    fi
fi

done

echo ""
echo "====="
echo "🚀 MISE À JOUR TERMINÉE SUR TOUT LE LAB 🚀"
echo "====="
```

exécution du script

```
root@serveurRAID:~# /root/update_all.sh
=====
🚀 DÉBUT DE LA MISE À JOUR GLOBALE 🚀
Date : mer. 24 juin 2026 08:42:41 CEST
=====

📦 -> Connexion à la machine : 192.168.15.254
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Atteint : 2 http://deb.debian.org/debian trixie-updates InRele
```

```
root@serveurRAID:~# /root/update_all.sh
=====
🔓 DÉBUT DE LA MISE À JOUR GLOBALE 🔓
Date : mer. 24 juin 2026 08:42:41 CEST
=====

📦 -> Connexion à la machine : 192.168.15.254
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Atteint : 2 http://deb.debian.org/debian trixie-updates InRelease
Ign : 3 https://build.openvpn.net/debian/openvpn/stable trixie InRelease
Ign : 4 https://download.webmin.com/download/newkey/repository stable InRelease
Ign : 3 https://build.openvpn.net/debian/openvpn/stable trixie InRelease
Ign : 5 http://security.debian.org/debian-security trixie-security InRelease
Ign : 4 https://download.webmin.com/download/newkey/repository stable InRelease
Ign : 3 https://build.openvpn.net/debian/openvpn/stable trixie InRelease
Ign : 5 http://security.debian.org/debian-security trixie-security InRelease
Ign : 4 https://download.webmin.com/download/newkey/repository stable InRelease
Atteint : 5 http://security.debian.org/debian-security trixie-security InRelease
```

```
(Lecture de la base de données... 207971 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../00-imagemagick-7-common_8%3a7.1.1.43+dfsg1-1+deb13u10_all.deb ...
Dépaquetage de imagemagick-7-common (8:7.1.1.43+dfsg1-1+deb13u10) sur (8:7.1.1.43+dfsg1-1+deb13u9) ...
Préparation du dépaquetage de .../01-libmagickcore-7.q16-10_8%3a7.1.1.43+dfsg1-1+deb13u10_amd64.deb ...
Dépaquetage de libmagickcore-7.q16-10:amd64 (8:7.1.1.43+dfsg1-1+deb13u10) sur (8:7.1.1.43+dfsg1-1+deb13u9)
Préparation du dépaquetage de .../02-libmagickwand-7.q16-10_8%3a7.1.1.43+dfsg1-1+deb13u10_amd64.deb ...
Dépaquetage de libmagickwand-7.q16-10:amd64 (8:7.1.1.43+dfsg1-1+deb13u10) sur (8:7.1.1.43+dfsg1-1+deb13u9)
Sélection du paquet linux-image-6.12.94+deb13-amd64 précédemment désélectionné.
Préparation du dépaquetage de .../03-linux-image-6.12.94+deb13-amd64_6.12.94-1_amd64.deb ...
Dépaquetage de linux-image-6.12.94+deb13-amd64 (6.12.94-1) ...
Préparation du dépaquetage de .../04-atril-common_1.26.2-4+deb13u1_all.deb ...
Dépaquetage de atril-common (1.26.2-4+deb13u1) sur (1.26.2-4) ...
Préparation du dépaquetage de .../05-atril_1.26.2-4+deb13u1_amd64.deb ...
Dépaquetage de atril (1.26.2-4+deb13u1) sur (1.26.2-4) ...
Préparation du dépaquetage de .../06-libatrilview3t64_1.26.2-4+deb13u1_amd64.deb ...
Dépaquetage de libatrilview3t64 (1.26.2-4+deb13u1) sur (1.26.2-4) ...
Préparation du dépaquetage de .../07-libatril-document3t64_1.26.2-4+deb13u1_amd64.deb ...
Dépaquetage de libatril-document3t64 (1.26.2-4+deb13u1) sur (1.26.2-4) ...
Préparation du dépaquetage de .../08-firefox-esr-l10n-fr_140.12.0esr-1-deb13u1_all.deb ...
Dépaquetage de firefox-esr-l10n-fr (140.12.0esr-1-deb13u1) sur (140.11.0esr-1-deb13u1) ...
Préparation du dépaquetage de .../09-firefox-esr_140.12.0esr-1-deb13u1_amd64.deb ...
Conservation de « détournement de /usr/bin/firefox en /usr/bin/firefox.real par firefox-esr »
Dépaquetage de firefox-esr (140.12.0esr-1-deb13u1) sur (140.11.0esr-1-deb13u1) ...

Progression : [ 17%] [ ]
```

```
✓ -> Mise à jour réussie pour 192.168.15.254

📦 -> Connexion à la machine : 192.168.15.253

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Ign : 1 http://deb.debian.org/debian trixie InRelease
Atteint : 2 http://security.debian.org/debian-security trixie-security InRelease
```

```

Found initrd image: /boot/initrd.img-6.12.90+deb13.1-amd64
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
✓ -> Mise à jour réussie pour 192.168.15.101

=====
  MISE À JOUR TERMINÉE SUR TOUT LE LAB
=====

```

configuration de MFA sur les different service

ssh

```

# Beware issues with some PAM modules and threads.
KbdInteractiveAuthentication yes

```

```

GNU nano 8.4 /etc/pam.d/s
@include common-session

# Print the message of the day upon success
# This includes a dynamically generated part
# and a static (admin-editable) part from /etc/motd
session optional pam_motd.so motd=/etc/motd
session optional pam_motd.so noupdate

# Print the status of the user's mailbox upon login
session optional pam_mail.so standard

# Set up user limits from /etc/security/limits.conf
session required pam_limits.so

# Read environment variables from /etc/environment
# /etc/security/pam_env.conf.
session required pam_env.so # [1]
# In Debian 4.0 (etch), locale-related environment variables are
# /etc/default/locale, so read that as well
session required pam_env.so envfile=/etc/default/locale

# SELinux needs to intervene at login time
# in the proper default security context.
# to run in the user's context should be run in the user's context
session [success=ok ignore=ignore module_unload=ignore]

# Standard Unix password updating.
@include common-password

auth required pam_google_authenticator.so

```

test connection avec MFA

```
root@ServerMaster:~# ssh root@192.168.15.254
The authenticity of host '192.168.15.254 (192.168.15.254)' can't be established.
ED25519 key fingerprint is SHA256:erTWYGc1NMNdi4GXwgYrHS4G+ZvrdPwTTG2J0onAF.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.15.254' (ED25519) to the list of known hosts.
(root@192.168.15.254) Password:
(root@192.168.15.254) Verification code:
Linux serveurRAID 6.12.90+deb13.1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.90-1
root@serveurRAID:~#

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@serveurRAID:~#
```

MFA au vpn

ajout du plugin MFA au server.conf

```
GNU nano 8.4 /etc/openvpn/server/server.conf *
keepalive 10 120
topology subnet
server 10.8.0.0 255.255.255.0
ifconfig-pool-persist ipp.txt
push "dhcp-option DNS 1.0.0.1"
push "dhcp-option DNS 1.1.1.1"
push "redirect-gateway def1 bypass-dhcp"
push "block-ipv6"
dh none
tls-groups X25519:prime256v1:secp384r1:secp521r1
tls-crypt-v2 tls-crypt-v2.key
crl-verify crl.pem
ca ca.crt
cert server_8rQeqQbNcj5oTFZF.crt
key server_8rQeqQbNcj5oTFZF.key
auth SHA256
cipher AES-128-GCM
ignore-unknown-option data-ciphers
data-ciphers AES-128-GCM
ncp-ciphers AES-128-GCM
tls-server
tls-version-min 1.2
remote-cert-tls client
tls-cipher TLS-ECDHE-ECDSA-WITH-AES-128-GCM-SHA256
tls-ciphersuites TLS_AES_256_GCM_SHA384:TLS_AES_128_GCM_SHA256
client-config-dir ccd
status /var/log/openvpn/status.log
management /var/run/openvpn-server/server.sock unix
verb 3
plugin /usr/lib/openvpn/openvpn-plugin-auth-pam.so openvpn
```


ajout de la case du mot de passe MFA (code 6 chiffre)

```
remote-cert-tls server
verify-x509-name server_8rQeqQbNcj5oTFZF name
auth SHA256
auth-nocache
cipher AES-128-GCM
ignore-unknown-option data-ciphers
data-ciphers AES-128-GCM
ncp-ciphers AES-128-GCM
tls-client
tls-version-min 1.2
tls-cipher TLS-ECDHE-ECDSA-WITH-AES-128-GCM-SHA256
tls-ciphersuites TLS_AES_256_GCM_SHA384:TLS_AES_128_GCM_SHA256
ignore-unknown-option block-outside-dns
setenv opt block-outside-dns # Prevent Windows 10 DNS leak
verb 3
auth-user-pass
```

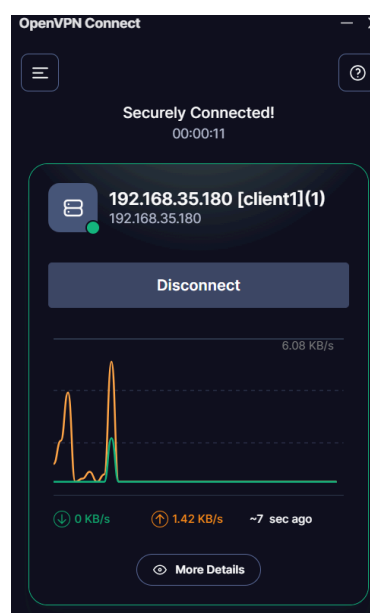
modification du fichier pam.d/openvpn

```
GNU nano 8.4 /etc/pam.d/openvpn
#@include common-auth
auth requisite pam_google_authenticator.so secret=/etc/openvpn/mfa/${USER}
account required pam_permit.so
```

creation du repertoire où le vpn pourra lire le fichier MFA

```
root@serveurRAID:~# mkdir -p /etc/openvpn/mfa
root@serveurRAID:~# chmod 700 /etc/openvpn/mfa
root@serveurRAID:~# cp /root/.google_authenticator /etc/openvpn/mfa/root
root@serveurRAID:~# chmod 400 /etc/openvpn/mfa/root
root@serveurRAID:~# nano /etc/pam.d/openvpn
```

test de connexion



MFA Idap

```
GNU nano 8.4 /etc/pam.d/common-auth
# (e.g., /etc/shadow, LDAP, Kerberos, etc.). The default is to use the
# traditional Unix authentication mechanisms.
#
# As of pam 1.0.1-6, this file is managed by pam-auth-update by default.
# To take advantage of this, it is recommended that you configure any
# local modules either before or after the default block, and use
# pam-auth-update to manage selection of other modules. See
# pam-auth-update(8) for details.
#
# here are the per-package modules (the "Primary" block)
auth [success=1 default=ignore] pam_unix.so nullok
# here's the fallback if no module succeeds
auth requisite pam_deny.so
# prime the stack with a positive return value if there isn't one already;
# this avoids us returning an error just because nothing sets a success code
# since the modules above will each just jump around
auth required pam_permit.so
# and here are more per-package modules (the "Additional" block)
# end of pam-auth-update config
auth required pam_google_authenticator.so secret=/var/lib/google-authenticator/${USER} nullok

root@serveurRAID:~# ssh client1@192.168.15.101
(client1@192.168.15.101) Password:
(client1@192.168.15.101) Verification code:
Linux client 6.12.94+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.94-1 (2026-06-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jun 25 10:47:12 2026 from 192.168.15.254
client1@client:~$
```

MFA sur webmin après avoir installer le module MFA ainsi que le compilateur perl

☆ Two-Factor Authentication

Enrolling for two-factor authentication with provider TOTP Authenticator ..
.. completed, with ID 3TXNXJZ0J5SFZV27

Enter the secret code 3TXNXJZ0J5SFZV27 in the TOTP app, or scan the QR code



Before logging out, you can test your new two-factor authentication setup here by
doesn't work, turn off two-factor authentication and try setting it up again.

Two-factor token

[← Retourner à la liste des utilisateurs](#)

☆ Two-Factor Authentication



Testing two-factor validation with TOTP Authenticator ..

.. test passed! You can now safely login using two-factor authentication.